

E-Commerce Customer Retention Analysis

Project Report

Tools Used: PostgreSQL | Power BI | DAX

Focus: Customer Retention | Cohort Analysis | Churn Behaviour

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2026

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1. Project Overview

This project looks at customer retention, churn behaviour, and revenue contribution for a mid-sized e-commerce business. The analysis was done entirely using SQL and Power BI, covering everything from raw data cleaning to final dashboard reporting.

The main questions driving this project were: How many customers actually come back after their first purchase? Which acquisition channels bring in better long-term customers? And how much of the revenue depends on retained customers versus one-time buyers?

The project was built as an end-to-end workflow, starting from PostgreSQL for backend analysis and finishing with an interactive Power BI dashboard for business reporting.

1.1 Project Objectives

- Analyse customer repeat purchase behaviour across segments
- Identify churn patterns broken down by acquisition channel
- Track customer retention trends over time using cohort analysis
- Segment customers based on their revenue contribution
- Evaluate whether discounts had any impact on repeat purchases
- Build an interactive Power BI dashboard to present findings

1.2 Tools and Technologies

Tool / Technology	Purpose
PostgreSQL	Database setup, data cleaning and SQL-based analysis
SQL	Writing queries, creating views and transforming raw data
Power BI	Building the interactive dashboard and visualisations
DAX	Creating calculated measures and KPIs inside Power BI
Data Modelling	Setting up relationships between the two datasets
Cohort Analysis	Tracking long-term retention patterns by signup month
Customer Segmentation	Classifying customers by purchase behaviour and revenue

2. Dataset Overview

Two datasets were used for this project. Both were imported into PostgreSQL and cleaned before any analysis was run.

Dataset	Description
customers.csv	Customer-level data including signup date, location and acquisition channel
orders.csv	Transaction-level data covering order dates, product categories, order values, discount status and order outcomes

The two datasets were joined on customer ID, which made it possible to analyse retention, churn and revenue patterns across different acquisition channels and time periods in a single workflow.

3. Data Cleaning and SQL Workflow

Before jumping into the analysis, both datasets were imported into temporary staging tables in PostgreSQL. Working through staging tables kept the raw data separate from the cleaned version, which made it easier to catch errors without affecting the final tables.

3.1 Data Import Process

The customer and order files were loaded into PostgreSQL as staging tables first. Once the data was validated and cleaned at that stage, it was pushed into the final analytical tables. This two-step import process is standard practice and helped avoid any messy data ending up in the queries.

3.2 Data Cleaning Steps

A few things needed fixing before the data was ready for analysis:

- Removed rows with blank or invalid customer IDs
- Used NULL checks to handle missing values across key columns
- Applied TRIM() to clean up inconsistent text formatting
- Converted date columns into proper PostgreSQL date format
- Cleaned order amount fields for accurate revenue calculations
- Filled in missing acquisition channel values where possible

After these steps, the dataset was clean enough to run reliable retention and churn analysis without worrying about bad records skewing the numbers.

3.3 SQL-Based Analytical Views

Rather than querying the raw transactional tables directly, four analytical views were created in PostgreSQL. These views act as a cleaner, business-ready layer that Power BI connects to for visualisation and KPI reporting.

View Name	Purpose
customer_segments	Classifies customers by purchase frequency into behaviour-based groups
churn_analysis	Flags customers as churned or active based on purchase recency
customer_value_segments	Groups customers by total revenue contribution
cohort_retention_rate	Tracks retention rates for each monthly customer cohort

4. SQL Logic and Analytical Workflows

4.1 Customer Segmentation Logic

Query 1 — customer_segments View

Customers were grouped based on how many orders they placed in total. The segmentation puts each customer into one of four buckets: No Purchase, One-time, Repeat, or Loyal. This classification feeds directly into the customer composition chart on the dashboard and forms the base for retention and revenue analysis.

```
CREATE VIEW customer_segments AS
SELECT
  c.customer_id,
  COUNT(o.order_id) AS total_orders,
  ROUND(COALESCE(SUM(o.order_amount),0),2) AS total_spent,
  CASE
    WHEN COUNT(o.order_id) = 0 THEN 'No Purchase'
    WHEN COUNT(o.order_id) = 1 THEN 'One-time'
    WHEN COUNT(o.order_id) BETWEEN 2 AND 4 THEN 'Repeat'
    ELSE 'Loyal'
  END AS customer_type
FROM customers c
LEFT JOIN orders o
  ON c.customer_id = o.customer_id
GROUP BY c.customer_id;
```

The LEFT JOIN ensures customers who never placed an order still appear in the output, which is important for tracking the No Purchase segment accurately. COALESCE handles null values in the order amount column so revenue calculations do not break for those customers.

4.2 Churn Analysis Logic

Query 2 — churn_analysis View

Churn was defined using a simple recency rule: any customer whose last order was more than 90 days ago is marked as Churned. Everyone else is Active. This is a common industry approach for e-commerce businesses and gives a clear, replicable definition that can be updated as the business evolves.

```
CREATE VIEW churn_analysis AS
SELECT
  customer_id,
  MAX(order_date) AS last_purchase_date,
  CASE
    WHEN MAX(order_date) < CURRENT_DATE - INTERVAL '90 days'
    THEN 'Churned'
```

```

        ELSE 'Active'
    END AS churn_status
FROM orders
GROUP BY customer_id;

```

The 90-day threshold was chosen because it reflects a realistic purchase cycle for a general e-commerce business. In a live project, this threshold would typically be calibrated based on the actual average order frequency of the customer base.

4.3 Cohort Retention Analysis

Query 3 — Cohort Retention Query

For the cohort analysis, each customer was assigned to a cohort based on the month of their first order. The query then counts how many unique customers from each cohort were still placing orders in the months that followed. This gives a month-by-month picture of how retention decays over time after acquisition.

```

WITH customer_cohorts AS (
    SELECT
        customer_id,
        DATE_TRUNC('month', MIN(order_date)) AS cohort_month
    FROM orders
    GROUP BY customer_id
)
SELECT
    cc.cohort_month,
    EXTRACT(
        MONTH FROM AGE(
            DATE_TRUNC('month', o.order_date),
            cc.cohort_month
        )
    ) AS month_number,
    COUNT(DISTINCT o.customer_id) AS active_customers
FROM orders o
JOIN customer_cohorts cc
    ON o.customer_id = cc.customer_id
GROUP BY
    cc.cohort_month,
    month_number
ORDER BY
    cc.cohort_month,
    month_number;

```

The CTE handles the cohort assignment by finding each customer's earliest order date and truncating it to the month level. The main query then joins back to the orders table to measure

activity in subsequent months, using AGE () and EXTRACT() to calculate how many months have passed since the cohort month.

5. Dashboard Development

The Power BI dashboard was built around a specific goal: telling the retention story in a way that makes sense to a business audience, not just a technical one. Instead of putting all the charts on a single page, the dashboard was split into four separate pages, each focused on one part of the analysis.

5.1 Dashboard Structure

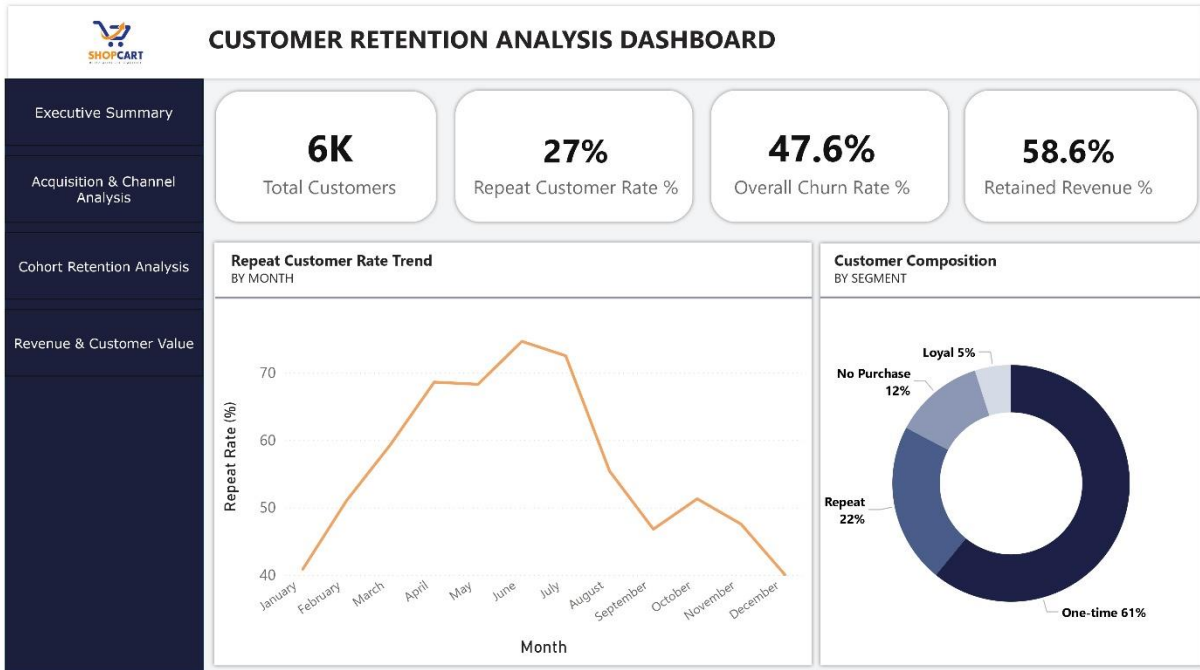
Page	What It Covers
Executive Summary	Top-level KPIs: repeat rate, churn rate, retained revenue percentage and monthly repeat customer trend
Acquisition and Channel Analysis	Churn and repeat rates broken down by acquisition channel and product category
Cohort Retention Analysis	Month-by-month retention heatmap for each customer cohort
Revenue and Customer Value	Revenue split by customer segment and the impact of discounts on repeat behaviour

5.2 Dashboard Design Approach

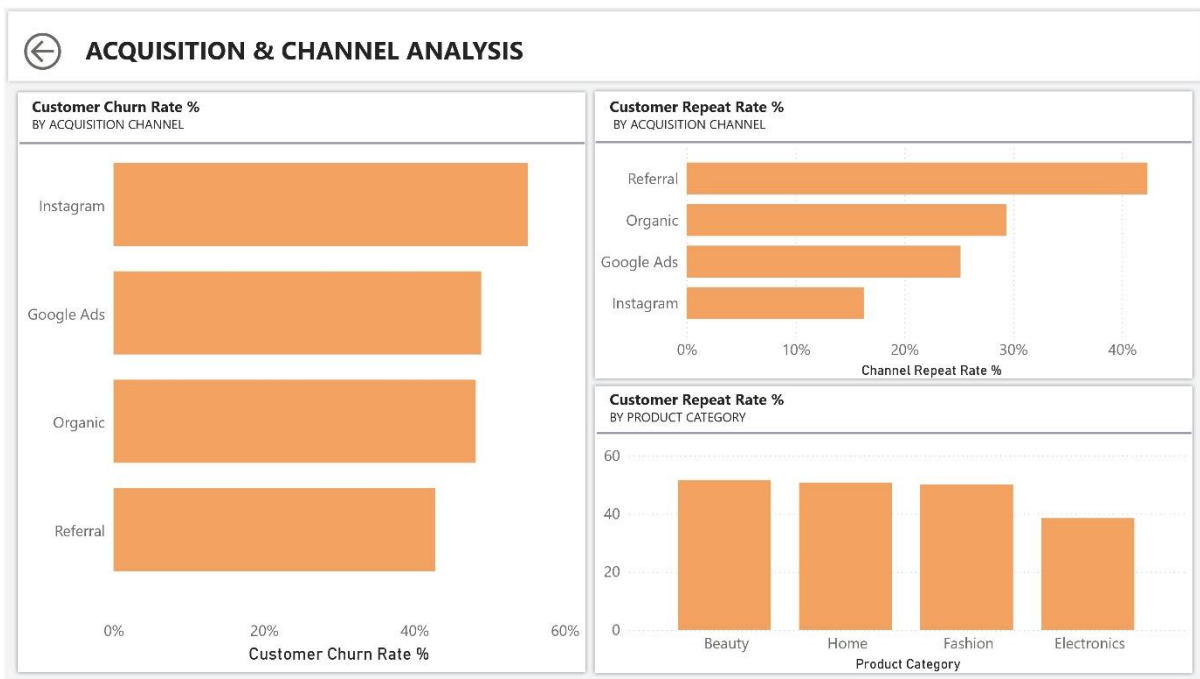
The design was kept intentionally simple. No unnecessary visuals, no colour overload. The goal was to make it easy for anyone looking at the dashboard to find the key number they need without having to hunt for it. A consistent layout and limited colour palette were used across all four pages to keep it looking like a single, coherent report rather than four disconnected slides.

5.3 Dashboard Screenshots

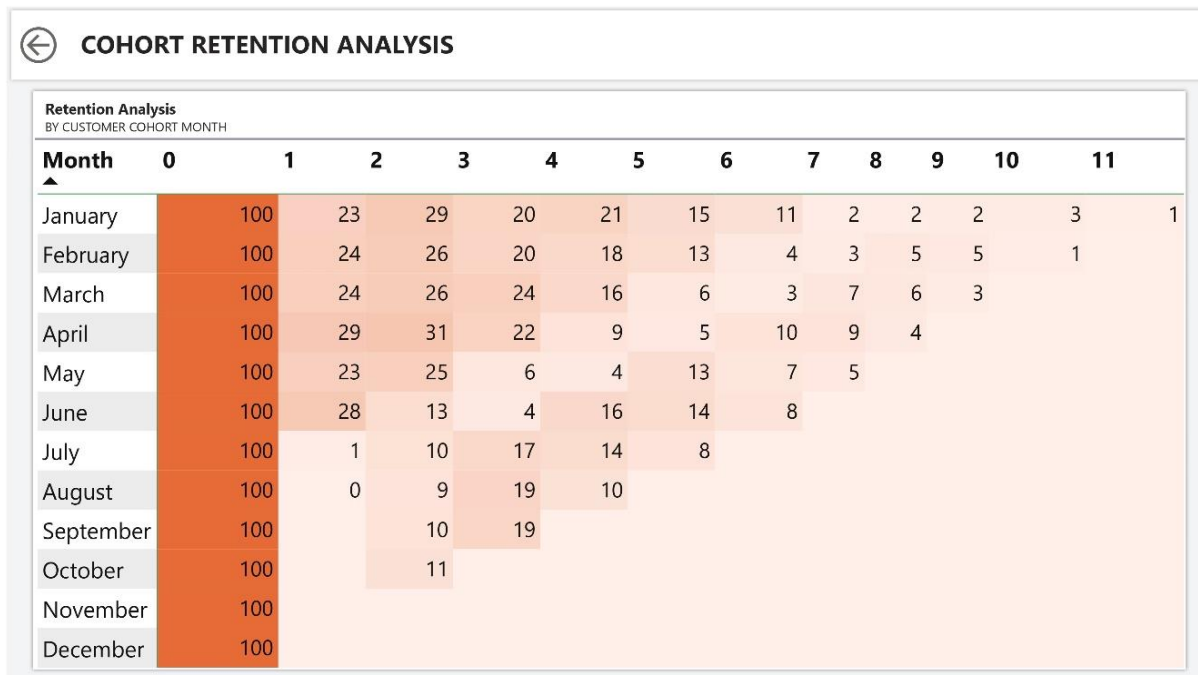
Executive Summary



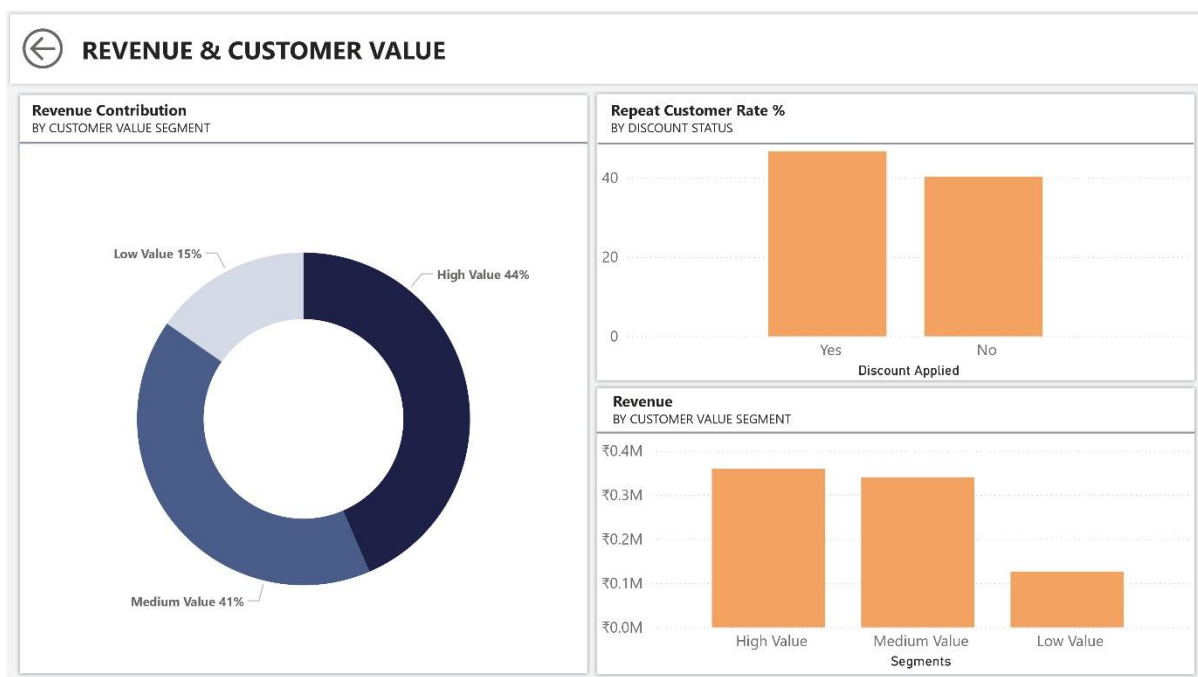
Acquisition and Channel Analysis



Cohort Retention Analysis



Revenue and Customer Value



6. Key Business Insights

After running the full analysis, a few patterns stood out clearly across retention, acquisition channels and revenue contribution.

Insight 1: Repeat Rate Dropped Sharply Over the Year

The repeat customer trend showed noticeable volatility throughout the year. Repeat rates started at around 42% in January, rose during the middle months, and then declined back to nearly 40% by December. At the customer level, 61% of users made only a single purchase, while just 5% reached loyal customer status. Overall, the analysis suggests that keeping customers engaged after their first purchase is the biggest retention challenge for the business.

Insight 2: Referral Was the Strongest Acquisition Channel

Referral-acquired customers had a repeat rate of around 42%, which was the **highest** of all four channels. Instagram came in at just 18%. That is a 24-percentage point gap, which is large enough to make a real difference in lifetime value and overall retention metrics. Referral customers also had the **lowest** churn rate, suggesting they tend to be more genuinely interested in the brand rather than just responding to a one-time ad.

Insight 3: Instagram Had the Highest Churn

Instagram-acquired customers had a churn rate of approximately 57% and a repeat rate of only 18%, making it the weakest channel on both metrics. Google Ads and Organic both sat around 42% to 43% churn, which is still high but noticeably better than Instagram. The pattern suggests that Instagram campaigns were bringing in a lot of **one-time buyers** rather than customers with **genuine long-term interest**.

Insight 4: A Small Segment of Customers Drove Most of the Revenue

High-value customers contributed **44% of total revenue** (approximately Rs. 0.35M) despite being a smaller portion of the customer base. Medium-value customers added another 41% (approximately Rs. 0.31M). Together, these two segments account for **85%** of total revenue. Low-value customers, despite being the most common type, contributed only 15% (approximately Rs. 0.12M). This kind of imbalance is fairly typical in e-commerce, but it does highlight how much the business depends on a relatively small group of loyal buyers.

Insight 5: Cohort Retention Fell Off Quickly After Acquisition

Across every cohort, retention dropped steeply in the first three months. The January cohort went from 100% to 23% by Month 1 and was down to 20% by Month 3. The April cohort had the best early numbers (29% in Month 1, 31% in Month 2) but still fell to just 4% by Month 8. Most cohorts dropped below 10% by Month 5 or 6. The pattern is consistent enough to suggest that the problem is not a specific campaign or period but a general gap in post-purchase engagement.

Data Quality Note: The July cohort showed an unusually low Month 1 retention of just 1%, which does not fit the pattern seen in any other cohort. This is most likely a data quality issue within the synthetic dataset rather than a real business event. In a live project, this kind of anomaly would be flagged and investigated before the report goes to any stakeholder.

Insight 6: Discounts Helped, but Not Dramatically

Customers who received a discount had a repeat rate of around 46.5%, compared to 40% for those who did not. That is a 6-percentage point difference. Discounts do seem to nudge customers toward a second purchase, but the gap is not wide enough to justify heavy discounting across the board, especially without knowing the margin impact. Targeted use of discounts on at-risk customers would make more sense than applying them broadly.

7. Business Recommendations

Recommendation 1: Focus More on Retention, Less on Acquisition

With a 47.6% overall churn rate and 61% of customers never coming back for a second purchase, spending more on acquiring new customers without fixing retention will not improve the situation. The business would get better returns by investing in post-purchase engagement such as loyalty programs, personalised follow-up emails and targeted re-engagement campaigns for customers who have gone quiet.

Recommendation 2: Shift Budget Toward Referral and Organic

Referral customers have a 42% repeat rate while Instagram sits at 18%. Before spending more on Instagram, it is worth understanding what makes referral customers stick around longer. Building a structured referral incentive program could help scale what is already working. For Instagram, either tighten the audience targeting or pull back spend until the campaign strategy is improved.

Recommendation 3: Fix the Drop-Off in the First Three Months

The cohort data shows that the steepest retention drop happens in Months 1 to 3. An onboarding sequence built specifically for this window, such as a welcome offer, product recommendations based on first purchase, and a re-engagement trigger around the 60-day mark, could make a meaningful difference in long-term retention numbers.

Recommendation 4: Protect the High and Medium Value Segments

These two segments together drive 85% of revenue. It is worth identifying what distinguishes them early, whether that is the acquisition channel, the product category they bought first, or the frequency of early purchases, and using those signals to improve targeting. Keeping these customers happy and engaged should be the highest priority retention goal.

Recommendation 5: Use Discounts Strategically, Not Broadly

The 6-percentage point lift in repeat rate from discounts is real but modest. Applying discounts broadly across all customers would erode margins without a proportional return. A smarter approach would be to target discounts specifically at one-time buyers showing signs of going cold, where the nudge is more likely to convert them into repeat customers.

8. Conclusion

This project covered customer retention behaviour, churn patterns, acquisition channel quality and revenue contribution using SQL and Power BI as the primary tools.

The numbers tell a fairly clear story: only 27% of customers made a repeat purchase, the overall churn rate sits at 47.6%, and a large chunk of revenue depends on a relatively small group of high and medium value customers. The business has a real opportunity to improve profitability by shifting some focus from acquisition to retention.

From a learning perspective, this project helped build a solid understanding of how SQL-based analysis connects to business reporting. Writing the views, defining churn logic, building the cohort query and then presenting the findings through a dashboard gave a good picture of how the full analytics workflow operates in practice.